

Cooperative Route Planning for Fuel-constrained UGV-UAV Exploration

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Abstract—Heterogeneous UGV-UAV system can efficiently complete large-scale tasks such as exploration and rescue by collaborative operation. This paper proposes a route planner based on a dynamic aggregation strategy to select the time-optimal routes for function-constrained UGV and fuel-constrained UAV to reach all the mission points. To reduce the time consumption of waiting at the take-off point, UGV and UAV are inclined to rendezvous at the next meeting point by no-wait operations. We present a two-step routing strategy to address this cooperative problem. First, the Voronoi diagram and greedy set cover method are used to determine meeting points and task areas. Then, solving a traveling salesman problem to find a UGV path by A-star and Lin-Kernighan heuristic. After that, an adaptive large neighborhood search heuristic is proposed to address vehicle routing problem with recharge for UAV among multiple task areas. The simulation and real experiment results demonstrate that our approach can effectively generate a time-optimal path for UGV-UAV Cooperative exploration.

Keywords—UGV-UAV, Fuel-constrained, Route planning, Adaptive large neighborhood search, Cooperative exploration

I. INTRODUCTION

In the past decade, the heterogeneous UGV-UAV system has had a rapid development and cooperative aerial-ground vehicle has been widely adopted in many applications including reconnaissance, surveillance, and exploration [1]. Unmanned aerial vehicles (UAVs) have emerged as a promising exploration tool because of their advantages of low weight, small size, and good flexibility. However, due to their limited battery capacity, they can only be used in relatively small areas.

By coordinating with the UAV, the UGV can provide recharging support for large-scale exploration [2]. Meanwhile, UGV can detect drivable areas to realize autonomous driving on rough roads based on the aerial image acquired from the UAV. On the basis of the new cooperative mode, we research the combinatorial optimization of UAV and UGV to further

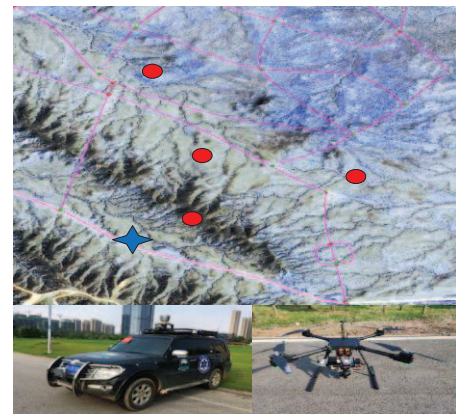


Fig. 1. An illustration of route planning for UGV-UAV exploration (red circle is the target point and blue star is the start point).

improve the efficiency of exploration. Fig. 1 shows an example scenario where there are lots of wide distributed target points that need to be arrived by fuel-limited UAV and UGV driving in the road network as a mobile recharging station.

This paper proposes suitable heuristics and a two-step optimization method to solve this non-deterministic polynomial-time problem in optimal time. The strategy starts by computing an iterative Voronoi diagram as an interpolation method in order to search the most suitable set of recharging stations for UAV relates to the road network. According to the result of partitioning, task points are allocated to the corresponding meeting point. Then, we determine the ideal number of meeting points using the greedy set cover method to cover every target point. Through domain decomposition, it is possible to reduce computing scale by turning a large-scale problem into several smaller discrete subproblems. On the one hand, an optimization method based on the A* algorithm reduces the distance traveled between meeting points by the UGV. On the other hand, the components of the adaptive large neighborhood search (ALNS) heuristic are tailored to solve UAV path planning preferred to generate routes across task areas. According to our knowledge, no previous work has been done on this sort of dynamic aggregation strategy.

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The rest of this paper is organized as follows: Section II gives a brief overview of the state-of-the-art development. Section III presents a mathematical formulation of the cooperative route planning problem and introduces the improved algorithm. Some experiments are conducted in real traffic situations and the results are presented in Section IV. Finally, we conclude this paper and discuss future research directions in Section V.

II. RELATED WORKS

Extensive research about cooperative route planning of fuel-constrained UGV-UAV has been carried out leading to a great many problem solutions. Chen J [3] classifies the Collaborative Methods of UGV-UAV systematically, applies the clustering method to target allocation, and discusses the multi-point dynamic aggregation problem. Tan C S [4] compares various route planning algorithms in terms of accuracy, time performance, and applicable environment.

Murray and Chu [5] put forward the FSTSP, which involves a single vehicle and a drone accomplish parcel delivery task together. A complementary problem to the FSTSP is PDSTSP which most delivery destination are located near the transportation route and stopping place. As a result, the drone is within flight range of near-neighboring customers, whereas the vehicle may independently service faraway consumers. Both problems were addressed with heuristics and mixed integer linear programming (MILP) and prove the effectiveness of the heuristic method. TSP-D formulation composed of trucks and drones share the same road network is proposed by Agatz et al. [6]. Additionally, drones can also be launched and returned to the same location, UAVs play a more effective role than before. Bouman et al. [7] extended the work by analyzing the problem from two perspectives. In order to solve the original TSP, the dynamic programming approach is used, which involves the shortest path algorithm applied to the TSP truck route and drone nodes. Moreover, they also presented an A* algorithm that is better in all cases for larger instances to solve the time performance of dynamic programming. Even so, the above methods only permit UAVs to visit one potential target point while flying, which negatively impacts the performance of the UAV in terms of time.

Maini and Sujit [8] define the Fuel Constrained UAV Refueling Problem with Mobile Refueling Station (FCURP-MRS). This work is similar to ours in that the UAV must visit each target site, but the mobile refueling station has no routing. On this basis, Ropero F [9] define Energy Constrained UAV and Charging Station UGV Routing Problem (ECU-CSURP). Cooperative routing for UGV-UAV considering fuel-constrain using cooperaTive ExploRation Routing Algorithm (TERRA) is proposed, but it just supports UAV-UGV start and end at the same point. When the scope of the task area is determined, UGV should stay at the meeting location until relevant UAV has visited all of the task points. Although this method can get the optimal total distance, it is not optimal in terms of the actual running time of the system. For the above reasons, we consider an extension of FCURP-MRS and TERRA to support dynamic aggregation which means UAV should consider adjacent areas and arrive at the next meeting point preferentially.

A branch and bound algorithm [10] and decomposition-based iterative heuristic [11] are proposed for improving solutions. propose a. There have been studies that have

reduced the number of optimization variables and constraints, enabling the exact methods to be computed more quickly. In addition to exact algorithms, heuristics algorithms such as Genetic algorithm [12] and Tabu search [13] show better performance as to solution quality, solving rapidity, and algorithm robustness for larger instances. The adaptive large neighborhood search (ALNS) heuristic has grown to be a well-known framework for applying heuristic solution techniques [14]. Ghilas V [15] proposed neighborhood-search-based heuristics to solve the Pickup and Delivery Problem with Time constraints and predetermine route. Results show that the ALNS is highly effective in finding good-quality solutions considering complex constraints. In this paper, we use a hybrid tabu search heuristic-based improved ALNS algorithm to find a feasible solution and improve the solution quality.

III. METHODS

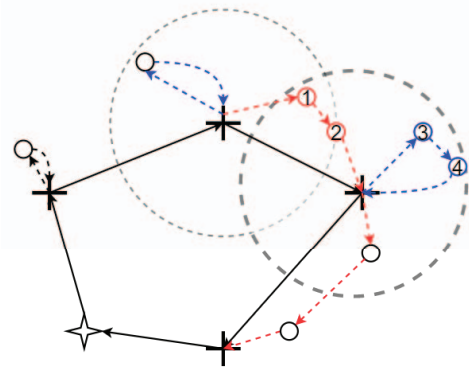


Fig. 2. An dynamic aggregation solution for ECU-CSURP

A. Problem Statement

The objective is to generate a cooperative routing for the heterogeneous UGV-UAV system to visit all targets while trying to minimize processing time. As soon as insufficient power is available to continue performing the next task for the UAV, it returns to the vehicle for battery replacement. As shown in Fig. 2, when several task nodes are more densely distributed and not time-consuming, the UAV may inspect several task nodes at a time before returning to the vehicle to replace the battery. In the process of returning, the UAV and the vehicle may rendezvous at the original meeting point, i.e., the vehicle is still parked at the original meeting point waiting after the UAV is deployed, such as point 3 and point 4 in Fig. 2; or the vehicle may also leave the original meeting point after the UAV is deployed, and the UAV and the vehicle rendezvous at another different meeting point, such as point 1 and point 2 in Fig. 2.

The basic assumptions are:

- UGV does not have any energy constraints.
- UGV and UAV move at fixed speeds, and UGV travels at a significantly slower speed than UAV.
- Not considering time spent at the meeting point for recharging and at the target point.

The sets and variables used to model the UAV multiple depot capacity vehicle routing problem are listed in Table I.

TABLE I. SETS AND VARIABLES USED IN THE FORMULATION

Sets and Variables	Description	Elements
V_M	The set of meeting points	$\{V_1, \dots, V_n\}$
V_T	The set of task points	$\{T_1, \dots, T_m\}$
E	The set of edges for UGV and UAV	$\{(i,j) i,j \in V\}$ $V = V_M \cup V_T$
x_{ij}	a binary variable judges UAV travel from i to j or not	$x_{ij} = 0$ or 1 $i, j \in V$
z_{ab}	a binary variable judges UAV travel from a to b or not	$z_{ab} = 0$ or 1 $a, b \in V_T$
w_{ij}	Binary variable judges UAV travel to j or return to i	$w_{ij} = 0$ or 1 $i, j \in V_M$

We express the FCURP-MRS with the following integer programming model:

$$\min \sum_{a,b \in V_T} \sum_{i,j \in V_M} z_{a,b} w_{i,j} \left(\begin{array}{l} \sum_{i \in V_M} x_{i,a} t_{i,a} + \\ \sum_{i,j \in V_T} x_{i,j} t_{i,j} + \\ \sum_{i \in V_M} x_{b,i} t_{b,i} \end{array} \right) \quad (1)$$

$$s.t. \sum_{i \in V} x_{i,j} = \sum_{i \in V} x_{j,i} = 1, \forall j \in V_T \quad (2)$$

$$\sum_{i,j \in V_M} x_{i,j} = 0 \quad (3)$$

$$\sum_{i \in V_M} x_{i,j} = \sum_{j \in V_T} x_{j,i} \quad (4)$$

$$\forall i \in V_M, \sum_{j \in V} x_{j,i} = 0 \wedge \sum_{j \in V_T} x_{i,j} > 0 \quad (5)$$

$$\sum_{i \in V_M} x_{i,a} t_{i,a} + \sum_{i,j \in V_T} x_{i,j} t_{i,j} + \sum_{i \in V_M} x_{b,i} t_{b,i} < T_{UAV} \quad (6)$$

$$x_{i,i} = 0, \forall i \in V \quad (7)$$

$$w_{i,i} = 1, \forall i \in V_M \quad (8)$$

$$x_{i,j}, z_{a,b} = 0 \text{ or } 1, \forall i, j \in V, \forall i, j \in V_T \quad (9)$$

Equation (1) is the objective function, which minimizes the total time cost for the exploration task including three parts. The first part involves the departure of the UAV from the meeting point and arriving at the task location. The second part is UAV travel among the task points. The Third part is the UAV returning to the meeting point for recharge. In this paper, the UGV path is preprocessed to generate a fixed value. The total time of cooperative exploration is the UGV time cost adding the UAV time cost.

Equation (2) implies the constraint that every task point should visit only once by the UAV. Equation (3) indicates the constraint that UAV don't need to travel along the UGV path. Equation (4) implies the constraint that UAV should land at UGV after completing the task. Equation (5) implies the constraint that the UAV can only start from a meeting point. Equation (6) represents the fuel constraints. Equation (8) shows that the role of w_{ij} is to judge UAV returns to the original meeting point or travels simultaneously with UGV to the next meeting point. If the UAV path is support dynamic aggregation, then the time cost for this path no longer counted in the total cost.

B. Solution Algorithm

In consideration of the solution space scale, we use a two-step optimization with a hierarchical struct to solve FCURP-MRS. Fig. 3 shows the whole system control scheme. Specifically, the algorithm consists of the following steps:

1) The Voronoi diagram and the greedy set cover algorithm are used to determine the necessary number of meeting points to cover all task points in the road network.

2) The A* and LKH algorithm is employed to plan the distance-optimal path for UGV along the road network. The hybrid ALNS algorithm is used to plan the time-optimal path for UAV.

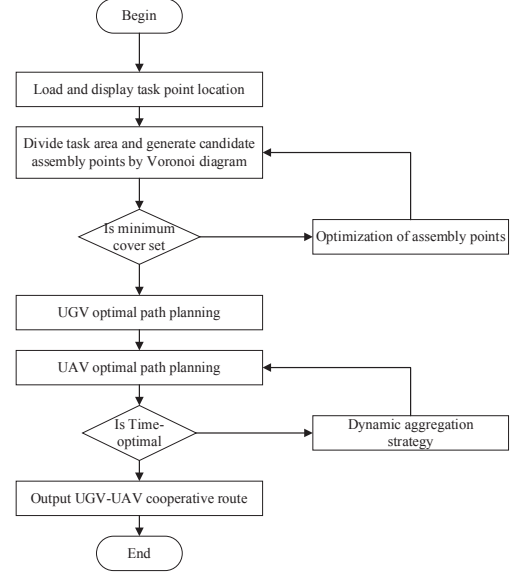


Fig. 3. The whole system control scheme.

C. First Step

The first step is comparable to the work in FCURP-MRS, placing intermediate UGV charging stops, so that, allows the UAV to reach its destination by replenishing its energy. However, our computational approach is founded on the boost polygon library to calculate the Voronoi diagram. The program generates an iterative Voronoi diagram that covers the desired locations in the road network. Map information is crucial to finding the correct meeting point. The topological struct of the road network is defined by nearest road curve and point. The method supports both points and edges as site types. The objective is to calculate a set of vertices located on the road network, whose areas cover all of the target spots. We tailored the implementation of a greedy set cover algorithm to search for a minimal set of vertices whose regions provide complete coverage of the collection of target points.

Algorithm 1 shows the main process. The inputs consist of the list of target sites and the UAV maximum flight time.

Algorithm1 Voronoi diagram and greedy set cover

```

1  Input:  $V_T$ 
2  Output:  $V_M$ 
3  //  $V_T^{nc}$ : set of target points not covered
4  //  $V_T^c$ : set of target points covered
5  //  $V_R$ : set of nearest road node in flying range
6  //  $V_{SCP}$ : set of alternative meeting points
7  While  $V_T^{nc} \neq \emptyset$  do
8    Compute Voronoi edges
9    Use map information to find  $V_R$  as  $V_{SCP}$ 
10   Connect the nearest task point as  $V_T^c$ 
11   Update  $V_T^{nc}$ 
12 end while
13 Get the  $V_T^c$  belong to  $V_{SCP}$ 
14 Order the set by weights using greedy strategy
  
```


D. Second Step

The goal is to determine the shortest path that stops at each vertex and travels back to the starting point. We solve for the UGV route using the traveling salesman problem formulation.

We preprocess map information and calculate the shortest distance among the meeting points using A* algorithm, then construct a distance matrix. It computes finite sub-tours for each attempting to get for the shortest distance traveled. We solve the above problem using LKH solver with Google's OR-Tools [16].

The UAV route can also be generated by OR-tools, but the result is not time-optimal. The main contribution is planning the route for UAV support dynamic aggregation strategy. We use heuristics local search in our problem to escape a local optimum solution. The proposed ALNS algorithm applies several neighborhood operators. By deleting some requests and reinserting them into the solution, a given set of possible routes is studied. Operators who performed well in earlier iterations had a better probability of being chosen than those who performed poorly. As a result, each operator is assigned a chance of being chosen. If the requirements given by the master search framework are met, the newly generated solution is accepted.

1) Initialization Stage

we used Google's OR-Tools solver to generate initial solution for (MDCVRP). Then, we used a set of destroying and repair methods in each iteration to find unfeasible and feasible solutions in large neighborhood spaces. In each iteration of the proposed method, just a few solutions are chosen to be destroyed and restored.

2) Probability Adjustment Procedure

To regulate the selection of operators, we employ a roulette-wheel mechanism. In each iteration, the weight is modified adaptively based on the performance of the destroy and repair procedures. We use the following formulation (10) to update the score and evaluate the probability of selecting a destroy-repair strategy.

$$P_d^{t+1} = P_d^t(1 - r_p) + r_p S_i \quad (10)$$

3) Destroy and Repair Stage

In this stage, we consider the influence of the next UGV position and add the trip-related removal. Remove the trips when we find the trip contains the repeated meeting point. So that, the next meeting point can be selected with high priority. The removal list contains the other removal operators including Random removal (RR), Historical knowledge removal (HR), Proximity-based removal (PR), and Worst removal (WR). We use Tabu Search and greedy repair method to make the repair stage more effective.

Algorithm2 hybrid ALNS algorithm

```

1 Initialize solution by OR-Tools
2 While not timeout and iters with improved do
3   Select destroy and repair heuristics
4   Destroy solution using operator in removal list
5   Repair solution using tabu search and greedy strategy
6   Get new solution cost and local best solution
7   Update acceptance criterion parameters
8   Determine whether to accept the solution or not
9   if accept
10    Update the current objective solution
11 else
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12   Revert back to the last local solution
13   Check if we found the new global best solution
14   Calculate new score and Update weights
15 end while
16 Output the global best solution
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IV. RESULTS

We produced 10 task points at random in a certain area of $5 \times 5 \text{ km}^2$. Table II shows the relevant parameters of the experiments including the speed of UGV and UAV, and the refueling time.

TABLE II. RELEVANT PARAMETERS OF THE EXPERIMENTS

Parameter	Value
v_{UGV} (The speed of the UGV)	5 m/s
v_{UAV} (The speed of the UAV)	10 m/s
T_{UAV} (The refuel time of the UAV)	10 min
S (The size of the exploration area)	$5 \times 5 \text{ km}^2$

A. Results of Route Planning

There are ten task points in the exploration region. Fig. 4 shows the result of step 1. Blue diamond presents the meeting point where UGV stops and recharges for UAV. Three meeting points cover all of the task points and divide them into sub-task areas.

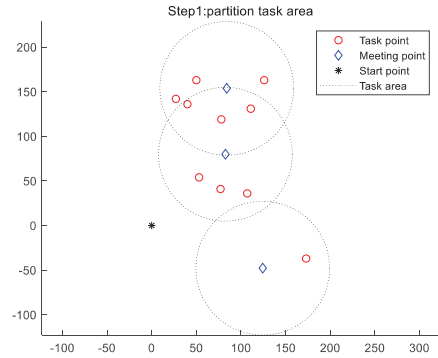


Fig. 4. Results for partition task area in step 1. Ten task points are divided into three circle areas.

Fig. 5 shows the simulation path planning results of UGV TSP. In a real experiment, we preprocess map information and calculate the shortest distance among the meeting points, then construct a distance matrix. The LKH heuristic method finds the final path along the road network.

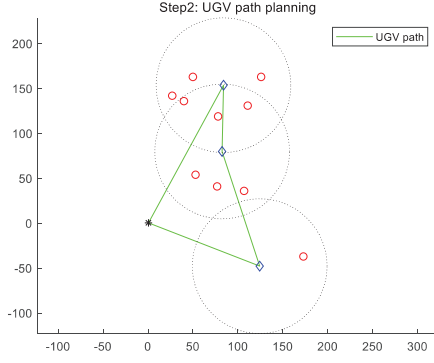


Fig. 5. UGV path generated by step 2. UGV starts from the start point and follows the green line to arrive at all of the meeting points.

Fig. 6 shows the UAV paths among task areas. UGV path sequence has been established in step 2. In the process of UAV route planning, UAV paths are generated based on MDCVRP formulation between the current area and adjacent area under the influence of UGV path direction. In the first task area, there are six task points belonging to one meeting point. In the adjacent area, there are three task points. There are overlapping parts between these adjacent areas and cover three task points. We can see that UAVs return to the original meeting point after visiting the part of task points and continue to complete the leftover task points while going to the next meeting point. Different from the TERRA result, the proposed method no longer cares about the task points in current region. The objective is to analyze the dynamic aggregation strategy performance in ECU-CSURP.

We apply TERRA method using GA and Optimal

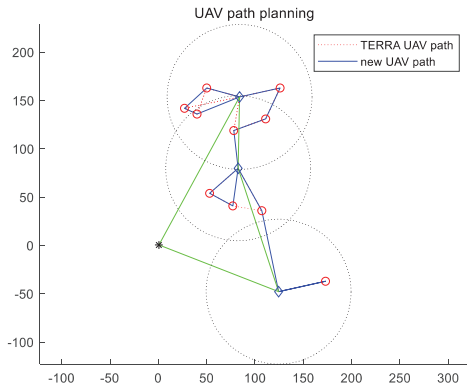


Fig. 6. Comparison UAV path between TERRA and the proposed method. The Red line represents the TERRA plan result and the blue line shows the new path result using the proposed method.

Operation Search Algorithm to generate the cooperative route with the same settings. The objective function values compare between TERRA and our proposed method validate the superiority of the dynamic aggregation strategy and hybrid ALNS algorithm to solve this problem. Table III shows the path sequence and object function value. Benefit from this dynamic aggregation strategy, the time cost is less than the TERRA result although the distance cost is longer. V_{index} presents the meeting point and T_{index} presents the task point. The most significant difference is TERRA path is planned in a separate area and ends like $V_1 - V_2$, otherwise

proposed path is plan in two adjacent areas and end like $V_2 - T_7 - V_3$. The last column contains the mission time, which is the time from start to finish. It can be noticed that the mission time decreases as the sequence of the UAV path is modified. In this case, we observed that the distance cost of the proposed method is 24.891km, but UGV could no-wait while UAV go with the path like $V_1 - T_6 - T_5 - T_4 - V_2$ and $V_2 - T_7 - V_3$. The result shows that time cost descend 37.2% than TERRA benefit from no-wait process.

TABLE III. THE OBJECTIVE FUNCTION VALUES OF THE DIFFERENT METHODS

Methods	Path Sequence	Object Function Value	
		Distance Cost (km)	Time Cost (s)
TERRA	$V_1 - T_1 - T_2 - T_3 - V_1$ $-T_4 - T_5 - T_6 - V_1 - V_2$ $-T_7 - T_8 - T_9 - V_2 - V_3$ $-T_{10} - V_3$	21.286	2128.6
Proposed	$V_1 - T_2 - T_3 - T_1 - V_1$ $-T_6 - T_5 - T_4 - V_2$ $-T_8 - T_9 - V_2 - T_7 - V_3$ $-T_{10} - V_3$	24.891 no-wait: 11.542	1334.9

B. Real Experiment

We develop a ground control station software based on Qt to realize cooperative route planning. Fig. 7 shows the process in the real experiment. The planner generates qualified global path points and transfer them to UGV-UAV through wireless networking communication module. Then, our self-driving vehicle outfitted with GPS/INS, a camera, and LiDARs can automatically drive on the rough roads following the preview points. UAV equipped with airborne optoelectronic pod can implement exploration tasks effectively. Experimental results prove the validity of our approach.

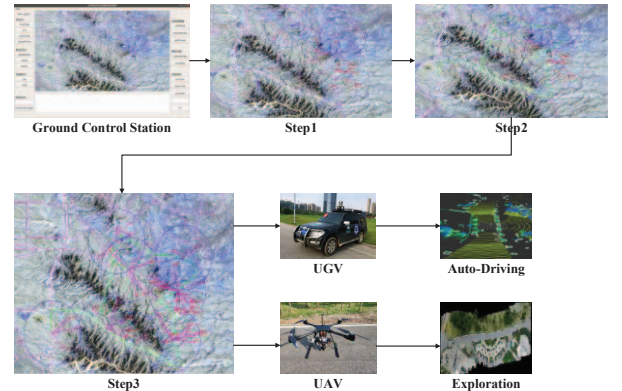


Fig. 7. The workflow of ground control station to plan global path for cooperative UGV-UAV exploration.

V. CONCLUSION

In this paper, we design a new formulation for the ECU-CSURP. The UGV can launch and recover the UAV from various positions on the road. The UAV can explore target points continuously within the longest endurance time, and

cooperate through effective path planning, to minimize the time wasted by UGV waiting for the UAV repeatedly return to the original point.

We proposed a two-step algorithm to meet the ECU-CSURP aims. First, it computes a Voronoi diagram to locate the recharge spots where the UGV will meet the UAV. Selects the optimal set of meeting points covering the full set of task points by greedy set cover algorithm. Then, it plans the UGV path using OR-Tools improved by A* and LKH algorithm. Finally, we improved ALNS algorithm with a new destroy operator called trip-related removal and tabu search as a repair method to find time-optimal UAV path.

The experimentation process aimed to evaluate the feasibility of the method and compare the total cost of TERRA and the proposed hybrid ALNS algorithm to solve the multiple depot capacitated vehicle routing problem. Our main observation is that the proposed method can effectively generate paths that support dynamic aggregation strategies. The cooperative route support UGV and UAV rendezvous at different meeting point considering the cross-task area to reduce time costs.

We can generate high-quality solutions for 42 mission points in less than two minutes on the ground control station. As the task points scale increases, the algorithm can produce a satisfying result in a short amount of time.

In future research, route planning of multiple UAVs and UGVs for cooperative exploration is an important direction to extend this work. Additionally, some critical variables and constraints in practical applications, such as hover time above the task point and terrain constraints need to be researched. The hybrid algorithm based on ALNS and intelligent algorithms is also worthy of in-depth study.

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